b) $CH_3 - \underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}$

c) $CH_3 - \underset{\displaystyle CH_3}{\overset{|}{CH}} - CH_2 -$

d) $CH_3 - \underset{|}{CH} - CH_2 - CH_3$

14. The number of stereoisomers of 1, 2 – dihydroxy cyclopentane

a) 1 b)2 c) 3 d) 4

15. Which of the following is optically active?
a) 3 – Chloropentane
b) 2 Chloro propane
c) Meso – tartaric acid
d) Glucose

16. The isomer of ethanol is

a) acetaldehyde b) dimethylether
c) acetone d) methyl carbinol

17. How many cyclic and acyclic isomers are possible for the molecular formula C_3H_6O?

a) 4 b) 5 c) 9 d) 10

18. Which one of the following shows functional isomerism?

a) ethylene b) Propane
c) ethanol d) CH_2Cl_2

19. $\overset{\ominus}{CH_2}-\underset{\underset{O}{||}}{C}-CH_3$ and $CH_2 = \underset{\underset{\overset{\ominus}{O}}{|}}{C} - CH_3$ are

a) resonating structure b) tautomers
c) Optical isomers d) Conformers.

20. Nitrogen detection in an organic compound is carried out by Lassaigne's test. The blue colour formed is due to the formation of.

a) $Fe_3[Fe(CN)_6]_2$
b) $Fe_4[Fe(CN)_6]_3$
c) $Fe_4[Fe(CN)_6]_2$
d) $Fe_3[Fe(CN)_6]_3$

21. Lassaigne's test for the detection of nitrogen fails in

a) $H_2N - CO - NH.NH_2.HCl$
b) $NH_2 - NH_2$. HCl
c) $C_6H_5 - NH - NH_2$. HCl
d) $C_6H_5 CONH_2$

22. Connect pair of compounds which give blue colouration / precipitate and white precipitate respectively, when their Lassaigne's test is separately done.

a) $NH_2 NH_2$ HCl and $ClCH_2–CHO$
b) NH_2 CS NH_2 and $CH_3 - CH_2Cl$
c) $NH_2 CH_2 COOH$ and $NH_2 CONH_2$
d) $C_6H_5NH_2$ and $ClCH_2 - CHO$.

23. Sodium nitropruside reacts with sulphide ion to give a purple colour due to the formation of

a) $[Fe(CN)_5 NO]^{3-}$
b) $[Fe(NO)_5 CN]^{+}$
c) $[Fe(CN)_5NOS]^{4-}$
d) $[Fe (CN)_5 NOS]^{3-}$

24. An organic Compound weighing 0.15g gave on carius estimation, 0.12g of silver bromide. The percentage of bromine in the Compound will be close to

a) 46% b) 34%
c) 3.4% d) 4.6%

25. A sample of 0.5g of an organic compound was treated according to Kjeldahl's method. The ammonia evolved was absorbed in 50mL of 0.5M H_2SO_4. The remaining acid after neutralisation by ammonia consumed 80mL of 0.5 MNaOH, The percentage of nitrogen in the organic compound is.

a) 14%
b) 28%
c) 42%
d) 56%

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